HWA CHONG INSTITUTION
JC2 Preliminary Examination
Higher 1

CANDIDATE NAME		**CT GROUP**	
CENTRE NUMBER		**INDEX NUMBER**	

GEOGRAPHY

8834/01

Paper 1

11 September 2024

INSERT

3 hours

READ THESE INSTRUCTIONS FIRST

This Insert contains all the Resources referred to in the questions.

Resource 1 for Question 1

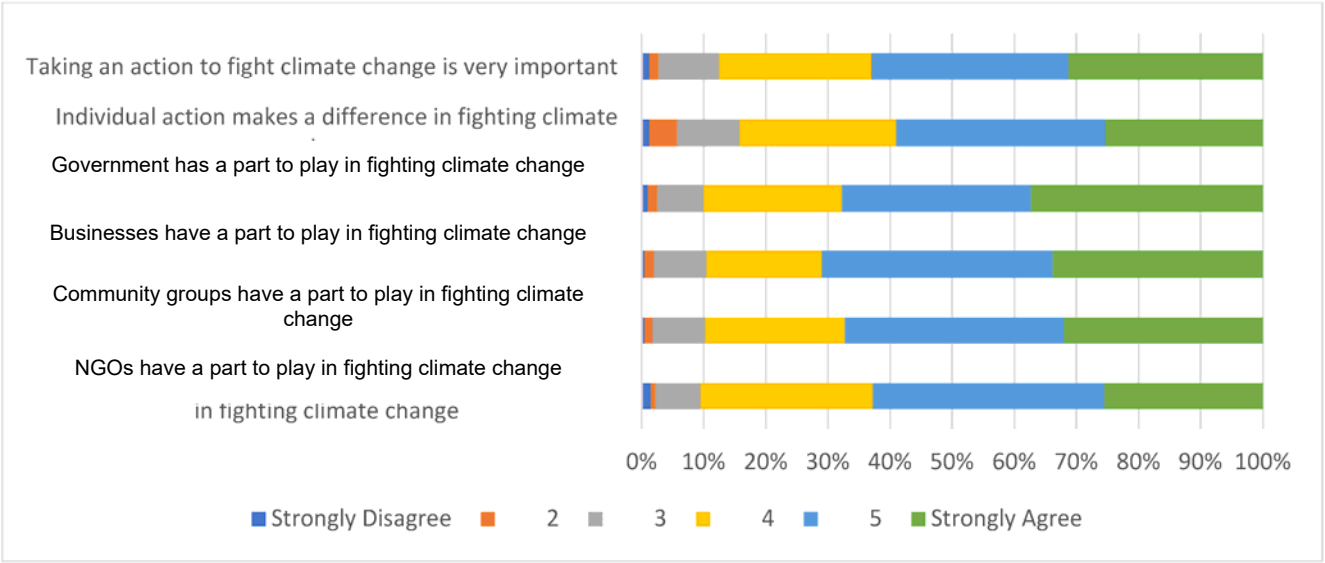
Street-level views outside Bugis Junction





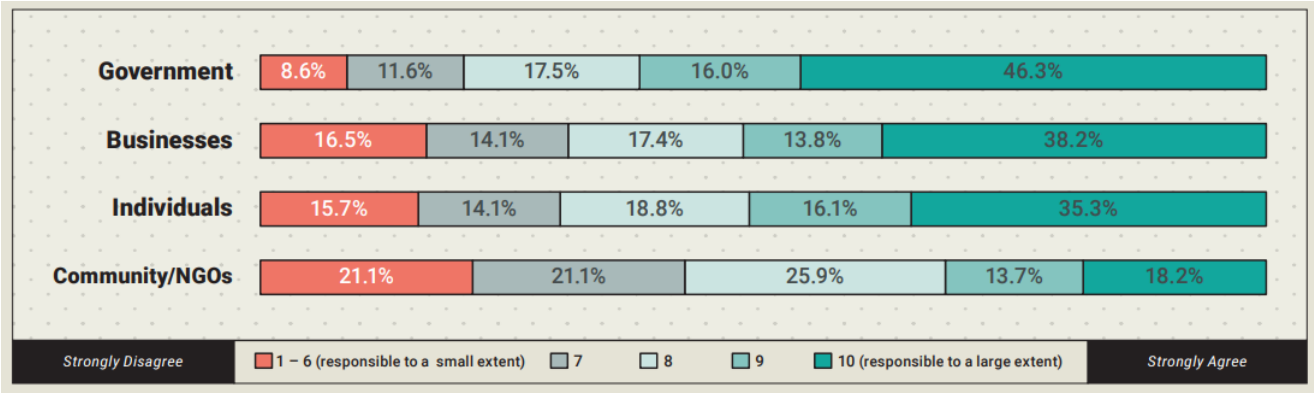
Resource 2 for Question 1

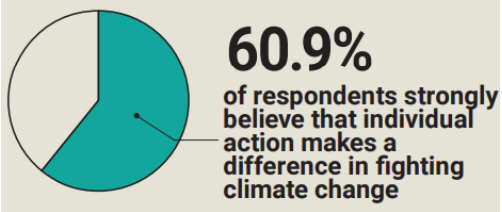
Surveyed Singapore consumers' perception of different actors' importance in fighting climate change

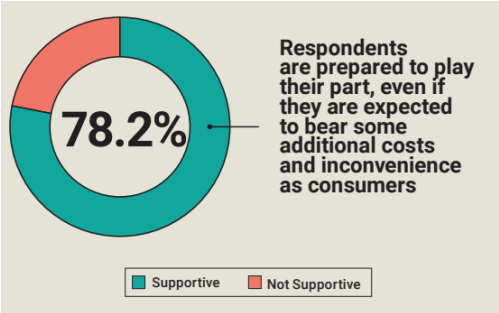


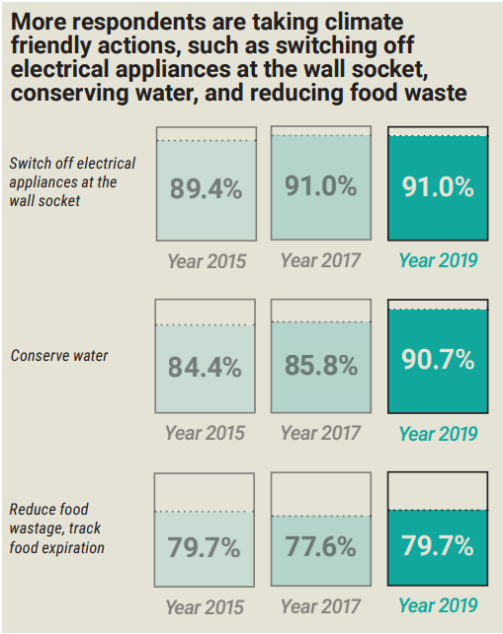
Resource 3 for Question 1

2019 selected survey findings by NCCS about Singapore consumers' perception on the responsibility of different actors in fighting climate change

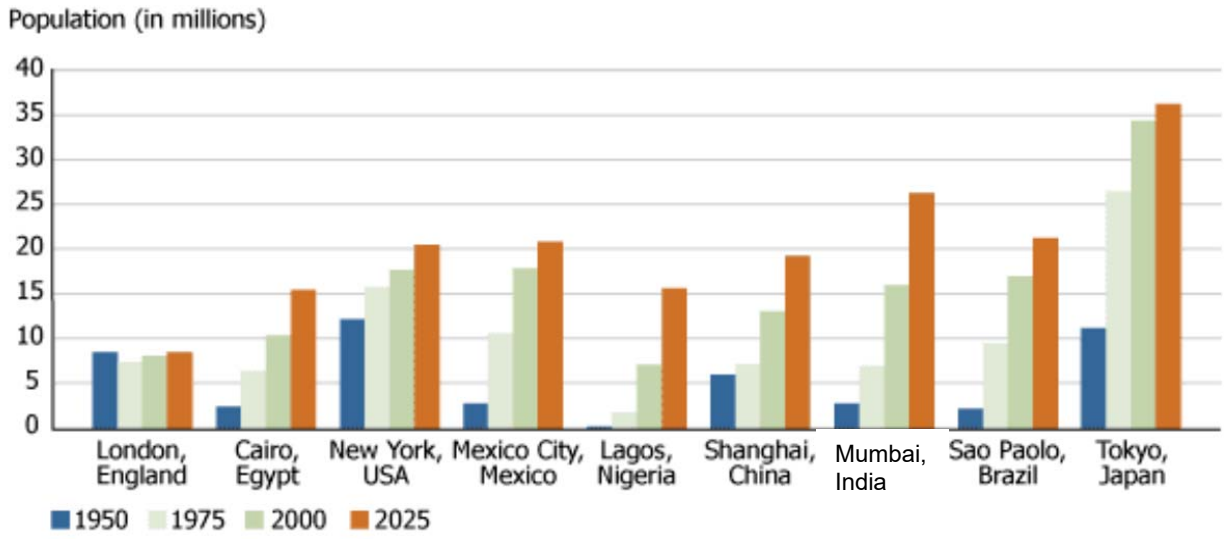






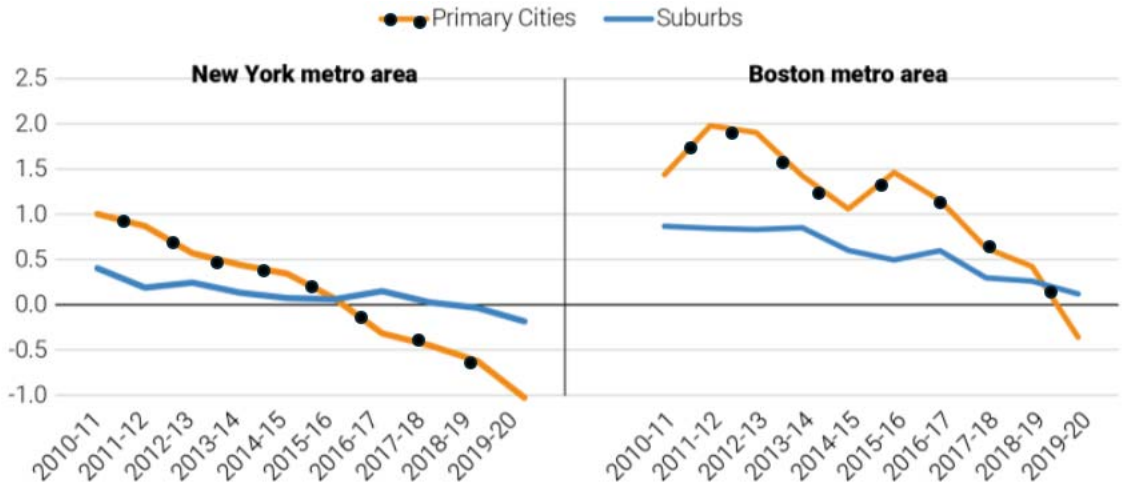


Resource 4 for Question 2
Urban growth in selected cities



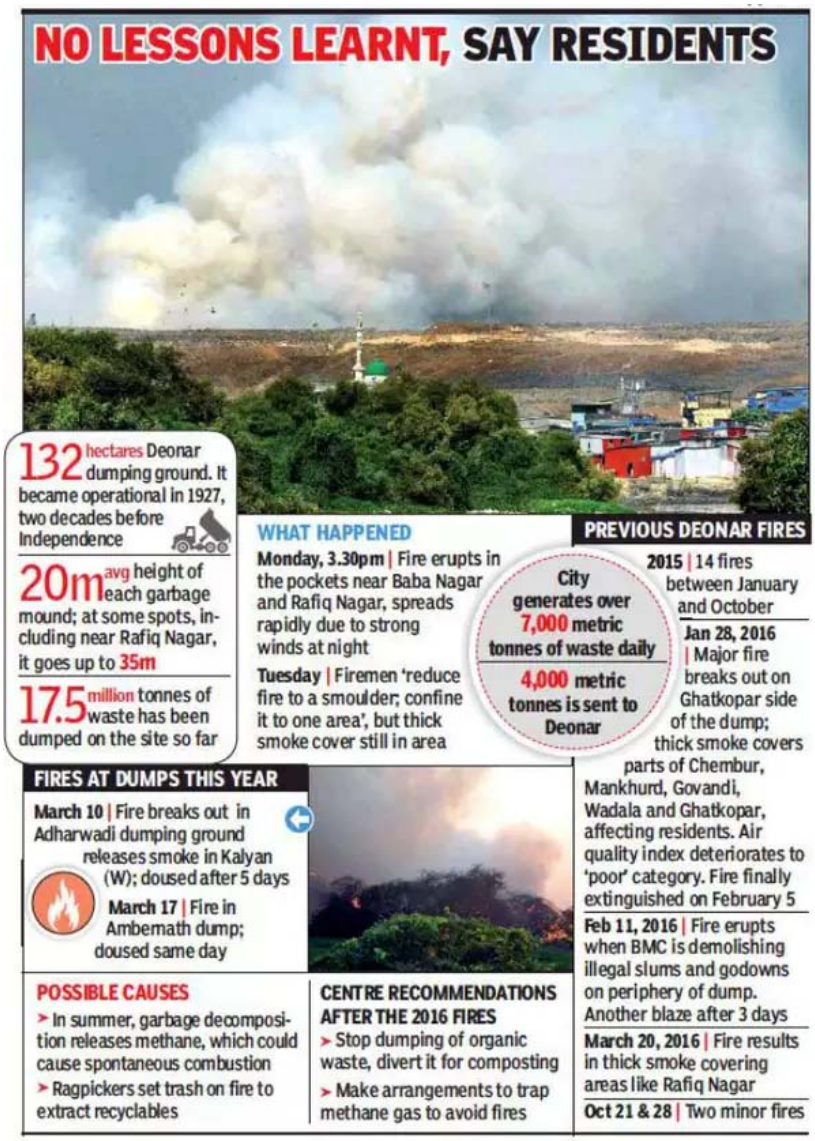
Resource 5 for Question 2

Urban population annual growth rates in primary cities and suburbs in New York and Boston metropolitan areas 2010 - 2020



Resource 6 for Question 2

Some details of the Deonar dumping ground fire in Mumbai, India and its location within the Mumbai Metropolitan Region





Major town in Mumbai

Resource 7 for Question 2

Solid waste production in Mumbai and Sweden and solid waste management by Tekniska Verken in Sweden

	Mumbai	Sweden
Area sq km	4,355	450,295
Population	20,748,395	9,851,017
Density (people/sq km)	4,764	22
Solid waste generated (tonnes/day)	7,025	5,879
Per capital waste (kg/person/day)	0.34	0.60

What Tekniska Verken does in Sweden each year

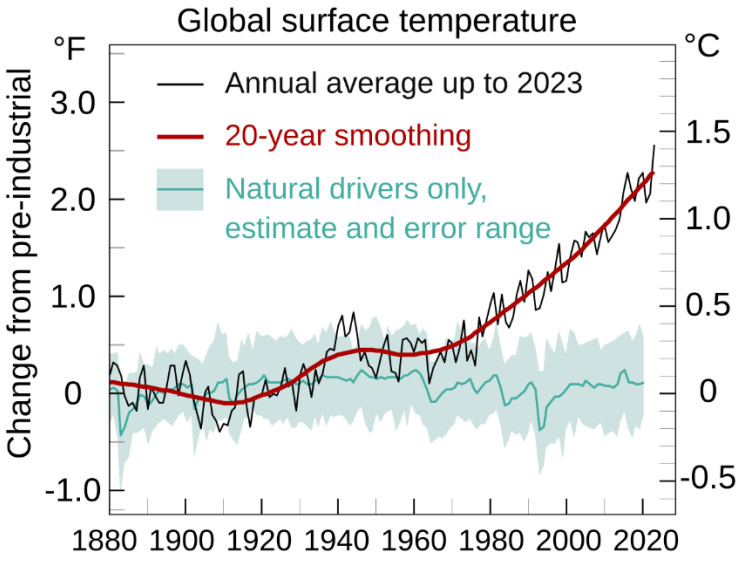
Currently converts 1,000,000 tonnes/year (2,739 tonnes/day) of waste into

•Biogas:	14 million Nm3 - 140 GWh
•Heat:	2,100 GWh
•Electricity:	680 GWh
300 000 regional customers in total - 14 000 district heating customers	
•Employees	about 1,000
•Sales:	575 million €
•Profit after net interest income and expense:	56 million €
•Net investment:	86 million €

* Tekniska Verken is a major Swedish multi-utility energy company
* Mumbai generates 5 025 tonnes of mixed waste (bio-degradable and recyclable)

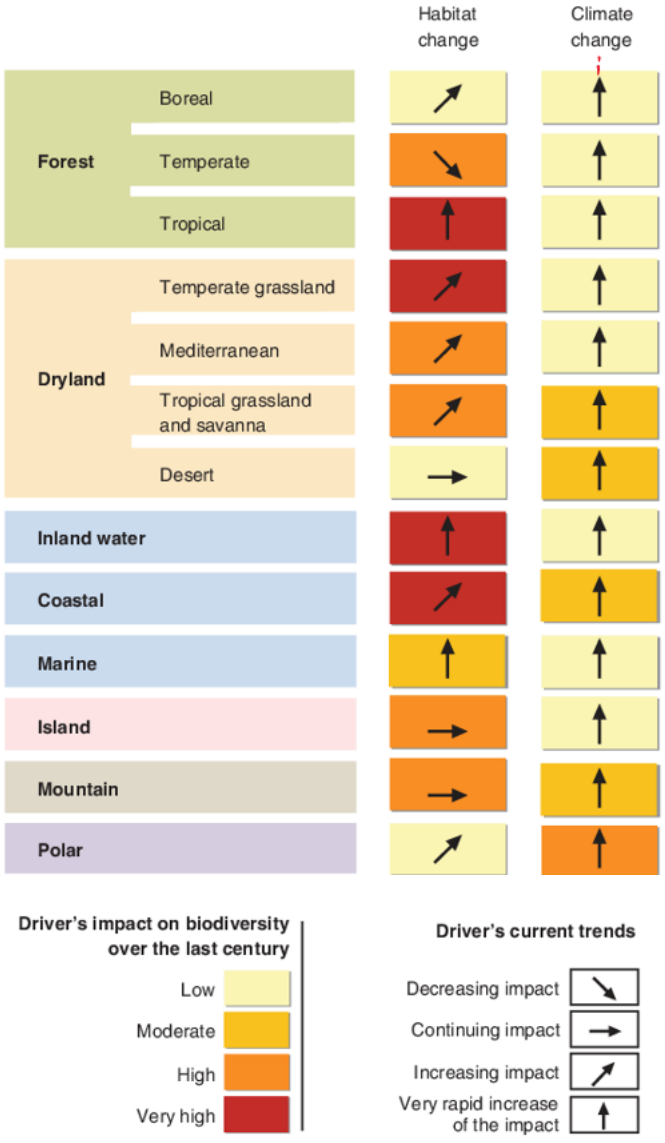
Resource 8 for Question 3

Changes in global surface temperature between 1880 to 2020



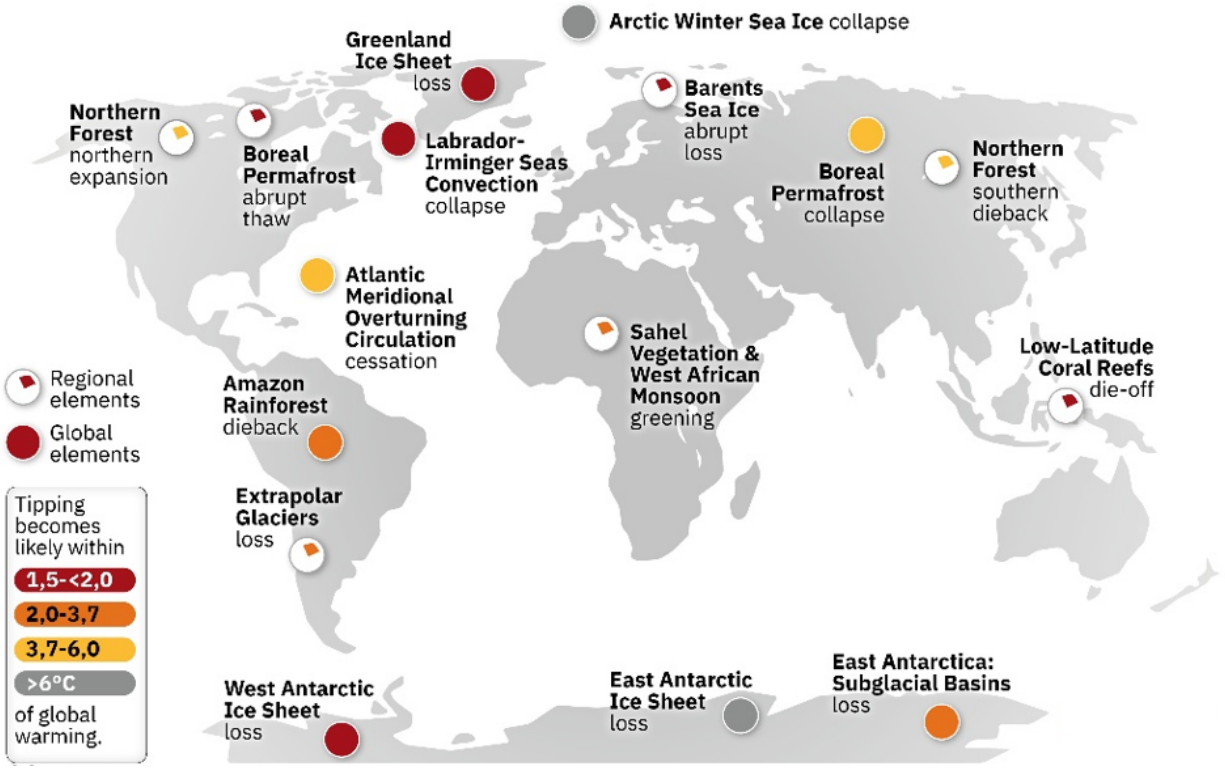
Resource 9 for Question 3

Effects of major drivers on biodiversity in Earth's ecosystems



Resource 10 for Question 3

Global warming and the tipping points of Earth's climate systems



* *Tipping points are 'critical thresholds in a system that, when exceeded, can lead to a significant change in the state of the system, often with an understanding that the change is irreversible*

Acknowledgements
Question 1 Resource 1 @ https://www.capitaland.com/sg/
Question 1 Resource 1 @ https://www.todayonline.com/singapore/
Question 1 Resource 2 @ https://www.mdpi.com/2071-1050/14/19/12330
Question 1 Resource 3 @ https://www.nccs.gov.sg/
Question 2 Resource 4 @ https://www.internetgeography.net/topics/what-is-urbanisation/
Question 2 Resource 5 @ https://www.brookings.edu/
Question 2 Resource 6 @ https://www.ibtimes.co.in/photos/
Question 2 Resource 7 @ https://asiaconverge.com/2016/03/669/
Question 3 Resource 8 @ https://www.ipcc.ch/report/ar6/wg1/figures/summary-for-policymakers
Question 3 Resource 9 @ https://www.semanticscholar.org/paper/
Question 3 Resource 10 @ https://www.pik-potsdam.de/en/output/infodesk/tipping-elements